PREDICTING LINEAR RELATIONAL DECODABILITY WITHOUT THE RELATION

Anonymous authors
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ABSTRACT

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations, including weak performance on PERSON_FATHER and stronger performance on COUNTRY_CAPITAL_CITY. We investigate the factors that contribute to this variation and demonstrate that it can be significantly predicted from a readout that is not aware of the underlying relationship. To do so, we propose a novel form of the Jacobian lens, the *contrastive entity lens*. Whereas the Jacobian lens is estimated across all tokens and concepts, the entity lens EL_t is trained to predict what the model will output for an entity of type t; e.g., we train ELs for countries and for persons. After estimating an entity lens, we measure how highly it ranks the model's answer for a relation – without using any relation-specific information in estimation and prediction. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.735$ ($p = 8.24 \times 10^{-6}$), using all 9,747 original rows, including demonstrations and references. We report the same full-evaluation comparison for six additional models. This association supports a role for information already carried by the subject representation, while leaving the causal mechanism open. At the fact level, generic first tokens and changes in the model's answer across demonstration contexts complicate interpretation. We report these limitations, the model-specific tokenization and prompt corrections, and readout baselines under the same full evaluation.

1 INTRODUCTION

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 87.5% of the time while one fitted on PERSON_FATHER agrees 7.5% of the time in our full evaluation.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Building on recent work by Gurnee et al. (2026), we introduce the *contrastive entity lens* EL_t, which is trained on subjects of type t placed in the neutral prompt `"{} is known for"` and evaluates the Jacobian at entities of type t. The construction changes what the readout is about: it is not about what a model might say in general about any concept, but about factual knowledge and, more specifically, about factual knowledge about an entity of a particular type. The question it answers – *what factual knowledge is this model ready to produce about this entity* – is specific to the entity type, but not specific to any relation.

We estimate the entity lens under the single neutral prompt `"{} is known for"` (shared by every relation), at one layer fixed across relations, and ask how highly it ranks the model's own answer. The lens sees no relation template, no objects and no (subject, object) pairs, yet its score correlates with LRE faithfulness across 28 relations at $\rho = 0.735$.

We make four contributions. (i) We introduce the entity lens, a supervision-free readout of what a model is disposed to say about an individual entity of a specific entity type. (ii) We turn it into a relation-agnostic diagnostic of linear decodability and show that it predicts LRE faithfulness. (iii) We evaluate seven models on complete relation datasets, compare readout baselines under the same protocol, and examine controls for object cardinality and answer-token frequency. (iv) Moving the unit of analysis from the relation to the individual fact, we provide an analysis of several problems in existing datasets – most prominently semantically weak first tokens and reference competition – that hinder further progress on understanding LREs in language models.

2 Background

An LRE for relation r approximates the model's computation as

$$o \;\approx\; \beta\, W s + b_\beta, \tag{1}$$

where s is the hidden state at the subject's final token and o the output state. Write f for the model's own map between the two: f sends the subject's hidden state at an intermediate layer ℓ to the final-layer state at the last position. We average local first-order approximations of f over a small set $\{s_i\}$ of reference subjects and scale their subject-dependent terms by β:

$$W = \mathrm{mean}_i\, J_i, \qquad b_\beta = \mathrm{mean}_i[\, f(s_i) - \beta J_i s_i \,], \tag{2}$$

where J_i is the Jacobian of f at s_i. At $\beta = 1$, this is the averaged first-order Taylor approximation (Hernandez et al., 2024).

This prediction decomposes into a subject-dependent term and a constant: expanding b_β gives

$$\beta W s + b_\beta \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i (s - s_i) \,], \tag{3}$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit β per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$.

Faithfulness is the rate at which decoding $\beta W s + b_\beta$ over the full vocabulary reproduces the *model's own* top-1 prediction (not the gold object). It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

3 Related work

Linear relational embeddings. Hernandez et al. (2024) brought the idea of modeling relations as linear maps between concept representations (Paccanaro & Hinton, 2000) to transformer language models (Vaswani et al., 2017). Rather than training such maps, they estimate them from a trained model by approximating the mapping from a subject's hidden state to the model's output with an affine function derived from the model's Jacobian. The approximation is faithful for some relations and poor for others. Merullo et al. (2025) trace part of this variation to pretraining data: LREs are more faithful when subjects and objects frequently co-occur in the corpus (we compare against such frequency measures in Appendix B.1). A more fundamental question is whether linear structure reflects what a model has learned at all, since many parametrizations induce the same next-token distribution. Marconato et al. (2024) refine the notion of relational linearity and show that, under suitable conditions, it holds either in all models that induce the same next-token distribution or in none. Valer et al. (2026) build on this formulation with an empirical probe based on the Kullback–Leibler divergence and find that relational linearity varies across models and layers and with how the relation is phrased. These works evaluate and explain linearity at the level of the relation. We ask instead how much of it can be predicted from how the model represents the subject, using a readout estimated without the relation or any of its objects. We also identify object cardinality as a second factor that shapes LRE faithfulness (§8).

Subject representations in factual recall. Mechanistic studies converge on the view that much factual knowledge is attached to the subject before the relation is processed. Feed-forward layers

operate as key–value memories whose keys respond to patterns in the input and whose values induce distributions over the output vocabulary (Geva et al., 2022), and causal tracing localizes factual associations in middle-layer MLPs at the last subject token, where targeted edits can change individual facts (Meng et al., 2022). Building on these findings, factual recall has been decomposed into stages: MLP layers enrich the subject representation with many of its attributes, the relation is propagated to the final position, and attention heads extract the requested attribute (Geva et al., 2023; Nanda et al., 2024). Nanda et al. (2024) further report that the lookup performed by early MLPs depends largely on the tokens of the subject's name rather than on the preceding context. Gottesman et al. (2026) show that knowledge acquisition tracks how often subject and answer co-occur, though frequency alone does not fully explain learning.

How facts are stored. Theoretical work offers two accounts of how transformers store facts. Nichani et al. (2025) model attention and MLP weights as associative memories and prove that a one-layer transformer can store a number of facts that scales linearly with its parameter count, with storage traded off between self-attention and MLP parameters. Ravfogel et al. (2026) propose a geometric alternative: in a controlled one-layer setting, subject embeddings encode linear superpositions of their associated attributes, and a small MLP acts as a relation-conditioned selector through ReLU gating rather than as a key–value lookup. The geometric account is close in spirit to our hypothesis. If an entity's attributes are already superposed in its representation, a relation only needs to select among them, and selection should succeed to the extent that the attribute is prominently encoded.

Lenses. Lenses read out intermediate hidden states in terms of the output vocabulary and differ mainly in the map they apply: the model's own unembedding (nostalgebraist, 2020), an affine translator learned per layer (Belrose et al., 2023), or the model's input–output Jacobian averaged over a large corpus and applied to raw activations, with the aim of identifying representations the model is poised to verbalize (Gurnee et al., 2026). Our contrastive variant evaluates the Jacobian at reference entities of the same type and applies it to differences between entity representations. This subtracts what the reference entities share, so that the top-ranked tokens describe an individual entity rather than its category. We compare the noncontrastive Jacobian lens under the same full-evaluation protocol (Appendix B).

4 THE ENTITY LENS

Gurnee et al. (2026)'s Jacobian lens is broadly similar to LRE. But instead of a relation-specific prompt and a focused target position (the object), it computes the average $W = \mathrm{mean}_i\, J_i$ over many diverse prompts and many output positions. The Jacobian lens also does not use the "anchor", the zero-order term of the Taylor expansion; so it is simply $W = \mathrm{mean}_i\, J_i$.

We derive our *contrastive entity lens* from Eq. 3, restated as:

$$f(s) \approx \beta W s + b_\beta \;=\; \mathrm{mean}_i[\, f(s_i)] + \beta\, \mathrm{mean}_i[\, J_i(s - s_i)\,] \tag{4}$$

Following Gurnee et al. (2026), we drop the anchor term $\mathrm{mean}_i[\, f(s_i)]$. We fix unit scaling for this readout and decode it with the model's final normalization and output head, each applied once. The final form of the contrastive entity lens is then:

$$\mathrm{CEL}(e) = \mathrm{mean}_i[\, J_i(s_e - s_i)\,] \tag{5}$$

CEL takes into account the difference between s_e and the reference point s_i – as it should if we want to approximate $f(s_e)$ using the first derivative. Our original motivation was that we are not interested in what the model associates with an entity like "Russia" in general (e.g., "Europe"). Instead we are interested in what the model associates with "Russia", but not with other reference entities like "Spain" and "Italy". Examples of such contrastive associations for "Russia" are "vodka" and "Kremlin". Our hypothesis was that such top-ranked contrastive associations give rise to linearity.

The subtraction makes the contrastive entity lens dependent on the reference set. For example, if s_e is a French person and a Spanish person is the reference entity s_i, then the vector difference in Eq. 5 can emphasize a direction associated with "French". If s_i is another French person, subtraction can instead suppress that shared direction, potentially affecting nationality and native-language readouts. This is a motivation for testing reference sensitivity, not a measured effect established by the present

correlations. We therefore also consider the following noncontrastive entity lens, which drops the reference set differences:

$$\mathrm{NEL}(e) = \mathrm{mean}_i[\, J_i s_e \,] \tag{6}$$

In what follows, EL refers to both NEL and CEL and "entity lens" refers to CEL unless further specified. The headline and cross-model results concern CEL; NEL is a separately measured baseline.

To instantiate the entity lens (for both NEL and CEL) for factual relations, we need (i) a prompt that evokes factual associations and (ii) a reference set of entities for computing the Jacobians J_i (both NEL and CEL) and for serving as the s_i in the vector differences $s_e - s_i$ (for CEL).

As the prompt we adopt `"{} is known for"`. This a neutral prompt, i.e., it is not customized for any specific factual relations. We use it for all relations, adding a leading newline for the models marked in Table 1.

As the reference set, we use entities of the same type as the subject, e.g., other countries if the subject is a country, other landmarks if the subject is a landmark etc. NEL applied to an input s outputs anything the model is likely to say in a context in which an entity of the same type as s occurs, including features it shares with other same-type entities. CEL outputs only what differentiates s from other same-type entities.

Computation. We do not compute J_i explicitly, but obtain each Jacobian–vector product $J_i u$ by the double-backward construction (Pearlmutter, 1994), at the cost of a constant number of forward and backward passes; this is what makes Eq. 5 affordable at $d_{\text{model}} = 4096$ (Appendix C).

The relation score. We decode $\mathrm{EL}(e)$ over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r, obtained under the relation's in-context prompt (§5). Writing $\mathrm{rk}_{\mathrm{EL}(e)}(m_e)$ for that one-based rank (the saved zero-based rank plus one), the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\, \log_{10} \mathrm{rk}_{\mathrm{EL}(e)}(m_e), \tag{7}$$

so higher A_r means the lens agrees with the model more.

What makes A_r relation-agnostic. Eq. 5 is not specific to r: the prompt is neutral and shared, the layer is fixed within each model (17 for OLMo; Appendix D) and no (subject, object) pair of the relation is used in estimation. Relation-specificity enters in exactly one place, the *target* m_e, the quantity faithfulness is defined against.

5 Experimental setup

Models and relations. We evaluate OLMo 3 7B, GPT-J 6B, Pythia 6.9B, Mistral 7B, Gemma 2 9B, Qwen3 8B Base and Qwen3.8 27B, using pinned checkpoints in `float32` with eager attention. OLMo is the main worked example. We retain the original 28 named-entity relations from the LRE dataset (Hernandez et al., 2024), comprising 9,747 rows, including duplicates. We do not expand to the other 19 relations or filter rows by correctness or first-token uniqueness. The model-specific year exception is documented in Appendix A.

Full evaluation. Each method retains its original reproducible selection of eight in-context demonstrations and eight reference rows. Every original row is evaluated, including demonstrations and references. The two methods use different seeded demonstration/reference selections; their evaluation multisets are identical, but their relation-context answer tokens can differ. The in-context prompt contains eight (subject, object) examples and a query subject without its answer. The lens uses this context only to obtain its target token; its hidden states and Jacobians always come from the neutral prompt.

For LRE, a six-layer by six-scaling grid selects the most faithful operator on the full evaluation. The scaling candidates are $\{0.5, 1, 1.5, 2, 3, 5\}$; the model-specific layer grids and fixed lens layers are listed in Appendix D. The lens omits self-reference comparisons for a query subject, but retains the query row in scoring. These are explicitly *in-sample* results: both reference/example reuse and LRE parameter selection involve evaluated rows. They do not estimate generalization to an independent held-out set.

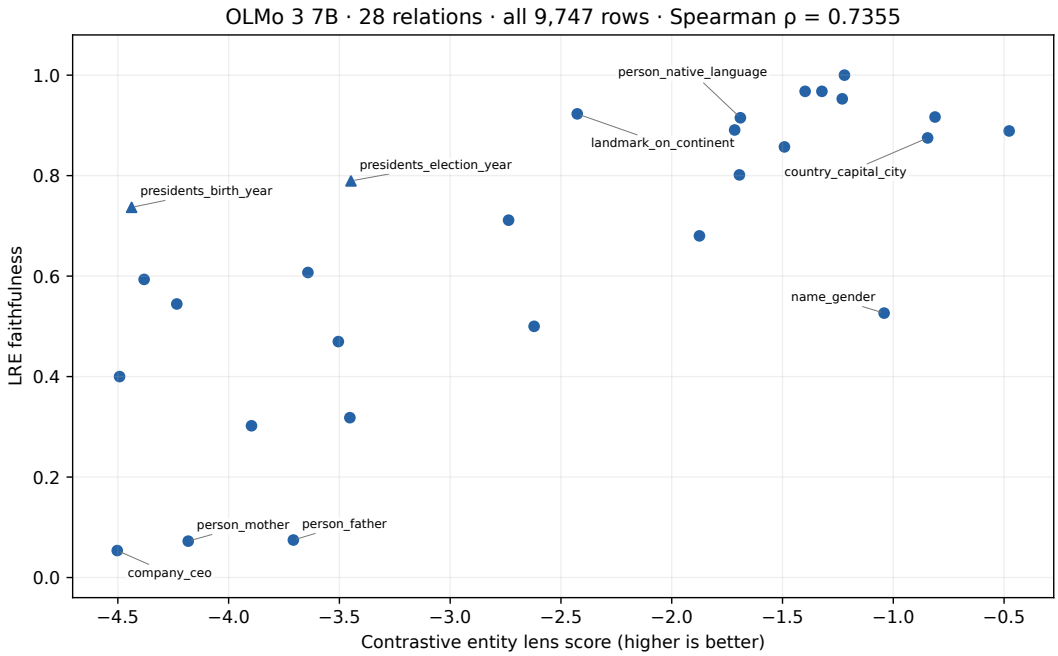


Figure 1: OLMo contrastive entity lens score against full-evaluation LRE faithfulness for all 28 relations and 9,747 rows. Every original row is included, with gender alternatives enabled and original continent references. Triangles mark the two year relations, retained after checking their token boundaries. Labels identify the residual extremes and relations discussed in the text.

Scoring corrections. Gender answers use the established same-gender alternative wording sets for lens ranks. LRE faithfulness remains exact agreement with the model's top-1 token. Continent references use the original seeded selection, without rebalancing. For GPT-J, Pythia and the two Qwen models, the neutral prompt starts with a newline: `\n{} is known for`. This avoids the large subject-state norms observed at the first token. The model names are marked in Table 1; Appendix A records all corrections and their scope.

6 Results

Figure 1 gives Spearman $\rho = 0.735$ ($p = 8.24 \times 10^{-6}$) for OLMo. Table 1 reports the same contrastive/differential readout for every model. The correlation is positive across the models but varies in magnitude. This is evidence that the relation-agnostic readout carries information about LRE faithfulness; it does not show that all models use an identical mechanism.

Controls. Across all 28 OLMo relations, cardinality has Spearman $\rho = 0.355$ with faithfulness ($p = 0.064$) and $\rho = 0.086$ with the lens score. The partial Spearman correlation controlling for cardinality is 0.757 ($p = 4.90 \times 10^{-6}$).

Answer-token prior has Spearman $\rho = -0.111$ with faithfulness and $\rho = 0.098$ with the lens score. Controlling for this prior gives $\rho = 0.755$; controlling for both prior and cardinality gives $\rho = 0.784$ ($p = 2.14 \times 10^{-6}$). The prior is the mean log probability of each evaluation target under ten neutral contexts, averaged over every original row. For partial correlations, we rank-transform both scores and each covariate, regress each score on the covariates with an intercept, and correlate the residuals. All controls use the complete 28-relation OLMo evaluation; no small-test-size cohort is selected.

Readout and frequency baselines. Appendix B compares the contrastive lens with the noncontrastive Jacobian lens, logit lens, differential logit lens and a Wikipedia attribute measure. Each comparison uses the same model-specific full evaluation and corrections. The contrastive lens does not uniformly outperform these baselines: the noncontrastive lens is stronger for several models, and

Table 1: Full-evaluation contrastive/differential Jacobian lens correlations with LRE faithfulness. LRE layer and scaling are selected on the full evaluation grid.

Model	Relations	Scored rows	Spearman ρ	p
OLMo 3 7B	28	9,747	0.7355	8.24×10^{-6}
GPT-J 6B†	28	9,747	0.4346	0.021
Pythia 6.9B†	28	9,747	0.5027	0.006
Mistral 7B	26	9,709	0.6561	2.74×10^{-4}
Gemma 2 9B	26	9,709	0.6889	9.96×10^{-5}
Qwen3 8B Base†	26	9,709	0.6647	2.12×10^{-4}
Qwen3.8 27B†	26	9,709	0.5539	0.003

†Leading newline in the neutral lens prompt to address high first-position activation norms. All eligible rows, including demonstration/reference rows, are evaluated. Models with 26 scored relations exclude the two year relations whose first answer token is the standalone digit 1; all 28 relations were measured. Gender alternatives are enabled; continent references use the original seeded selection.

ordinary logit lens has the largest point correlation among the vocabulary-rank readouts for GPT-J and Qwen3 8B Base. These comparisons do not establish statistically significant differences between readouts. OLMo corpus-frequency features are reaggregated over all original rows before correlation with the current faithfulness values.

7 Analysis of the entity lens

7.1 Neutral prompts and reference composition

The neutral prompt asks what an entity is known for. Its readout may therefore emphasize identity, profession or cultural associations while ranking gender, family members or university names poorly. We interpret this as a possible limitation of the probe, not proof that those facts are absent from the representation. The present results do not establish that salience is the causal mechanism behind LRE faithfulness.

7.2 Reference sets

The contrastive readout averages $J_i(s_e - s_i)$ over reference entities of the same subject type. If a queried subject and its references share an answer-related direction, subtraction can suppress that direction. The noncontrastive readout $\mathrm{mean}_i\, J_i s_e$ does not perform this subtraction; we compare it under the same full-evaluation protocol in Appendix B.

The current measurements retain the original seeded reference rows, including their answer-category imbalance. In particular, the landmark-continent reference sets are not balanced across continents. This makes the choice reproducible, but it does not make the references neutral with respect to every relation. Reference-composition robustness is not established by the present aggregate correlations.

8 Relation-level outliers

We fit a least-squares line to the OLMo points in Figure 1 and inspect residuals in units of faithfulness. The largest positive residuals are PRESIDENTS_BIRTH_YEAR ($+0.375$, $k = 1.3$), PRESIDENTS_ELECTION_YEAR ($+0.271$, $k = 1.1$), LANDMARK_ON_CONTINENT ($+0.242$, $k = 189.4$). The largest negative residuals are PERSON_FATHER (-0.403, $k = 1.0$), NAME_GENDER (-0.374, $k = 9.5$), PERSON_MOTHER (-0.330, $k = 1.0$). These residuals are recomputed from the least-squares fit to the full-evaluation points; they are measured in faithfulness units. These examples show why cardinality and lens agreement need not have the same association with faithfulness. Repeated answer tokens can make an operator faithful without fine-grained subject discrimination, whereas a large answer space makes exact top-1 agreement difficult. The residuals are descriptive; the full-data controls in Section 6 do not establish a causal explanation for either tail.

9 FACT-LEVEL ANALYSIS

We join all 9,747 OLMo evaluation rows by subject and object, retaining duplicate occurrences, and correlate the two answer-token ranks within each relation. Correlations are defined for 27 of 28 relations, ranging from -0.401 to 0.664, with median 0.235 among defined relations. Constant ranks make the statistic undefined for COUNTRY_LANGUAGE; their rows are retained in every other analysis. The two methods use different demonstration contexts, whose target tokens disagree on 2,115 of these rows. These within-relation associations therefore combine readout differences with context sensitivity; they do not compare an identical target under matched contexts. Appendix F lists every relation, including undefined correlations caused by constant ranks.

9.1 CONTEXT AND ENTITY AMBIGUITY

A subject string can evoke more than one entity, and the neutral prompt can emphasize associations other than the requested relation. Different in-context examples can also change the model's answer. Both issues complicate an interpretation of rank disagreement as evidence that a fact is missing from a linear representation. A future matched-context experiment should separate these effects while retaining the complete evaluation; the current protocol preserves the original contexts.

9.2 SEMANTICALLY WEAK FIRST TOKENS

The first token of a multi-token answer can be shared by many distinct objects. University names provide a concrete full-evaluation example: the selected GPT-J LRE predicts "University" for every one of the 91 rows and obtains 79.1% faithfulness; Pythia does the same and obtains 85.7%. These scores exactly match their constant majority-token baselines. Thus high faithfulness can reflect agreement on a generic prefix rather than recovery of a university's identity. This issue persists after the newline correction. We retain these rows and report the limitation instead of filtering away inconvenient objects or model predictions.

9.3 TOWARDS BETTER FACT EVALUATIONS

Answer scoring should distinguish complete objects rather than credit a shared, semantically weak first token. Complete-answer scoring is a separate metric and is not substituted into the present first-token faithfulness results. Likewise, evaluations should distinguish ambiguity in the intended subject from failure of the relational readout, and should report when their demonstration contexts induce different targets.

10 INTERPRETING THE OPERATOR'S TWO TERMS

The affine operator in Eq. 2 separates a subject-dependent term $\beta W s$ from an anchor b_β fixed by the references. This algebra motivates comparisons with neutral-prompt Jacobian readouts, but it does not show that a neutral readout can substitute for the subject term of a relation-context LRE. We therefore interpret the full-evaluation correlations as predictive associations. Establishing interchangeable terms would require an intervention study with explicit complete-data coverage and scoring, beyond the comparisons reported here.

11 LIMITATIONS

The benchmark covers seven models and 28 original relations, with two model-specific year exclusions in four models. It remains a small relation-level sample, and related datasets can share subjects and objects. Correlations and unadjusted p-values are descriptive evidence rather than an identification of causal mechanisms.

The evaluation includes demonstration and reference rows and selects LRE layer/scaling on the scored data. This improves coverage of small relations but creates in-sample dependence; the reported correlations must not be read as independent held-out performance. Gender has only 19 rows, including a duplicate, even under complete evaluation.

The neutral prompt, reference composition and fixed lens layer all influence the readout. Newline corrections were motivated by activation norm diagnostics and evaluated on these same relations. We do not claim an independent validation of their effect or robustness across arbitrary neutral prompts. First-token scoring can still reward generic prefixes, and different demonstration contexts can change the two methods' target tokens. Complete-answer and matched-context evaluation remain open work.

12 CONCLUSION

A contrastive Jacobian readout computed under a neutral prompt predicts variation in LRE faithfulness across complete relation evaluations. The association holds across seven models, with the largest first-token norm artifacts reduced by a leading newline where needed. Its magnitude and its outliers depend on the model and scoring protocol. Full-row coverage, explicit tokenization corrections and aligned baseline protocols are necessary for interpreting this evidence.

AI USE STATEMENT

We used generative AI tools for brainstorming, assisting with software code, and initial drafting and rewriting of parts of the manuscript. All AI-assisted code and text were reviewed and verified by the authors, and generated code was tested for correctness. The authors take responsibility for the final content of this work, including all text, claims, results, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

We provide sufficient details to reproduce all experiments in the main text and appendix, including the datasets, model checkpoints, prompt templates, evaluation procedures, and hyperparameters used throughout. All experiments use publicly available model checkpoints and datasets. We report the procedures used to estimate linear relational embeddings and the contrastive Jacobian lens, as well as the definitions of all faithfulness and ranking metrics. Additional implementation details, robustness analyses, and per-relation results are provided in the appendix. Code for reproducing the experiments and figures will be released upon publication.

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A FULL-EVALUATION PROTOCOL AND CORRECTIONS

All benchmark tables and figures in this paper use the original 28 relations, comprising 9,747 rows per model before the stated year-scoring exception. Original strings, labels, row order within the recorded selections, and duplicates are preserved. There is no correctness filter, identifying-first-token filter, row cap or minimum-test-size cohort.

Demonstrations, references and evaluation. Each method uses eight demonstration rows and eight reference rows. The LRE selection reproduces the original sequential shuffles with seeds 0 and 1000; the lens selection uses seed 17. Every original row is then evaluated, including both selections. For the lens, reference comparisons sharing the query's subject string are omitted from its average; the query is still scored, and duplicates remain in aggregate statistics. LRE retains its original reference estimator. The different demonstration contexts can produce different target tokens, which is recorded rather than silently treated as a matched-context evaluation.

LRE selection. The primary outcome chooses a layer and scaling from the original six-by-six grid on the full evaluation. Table 3 also reports the original fixed layer/scaling evaluated on every row. Neither version is an independent held-out estimate. Full-data LRE measurements and target tokens are reused for neutral-prompt changes; only affected lens readouts are recomputed.

Gender wording. For a target in the female wording set, the accepted single-token, leading-space terms are `woman, her, herself, she, hers, female, girl`; for the male set they are `man, him, himself, he, his, male, boy`. A lens rank is the best rank in the matching set; other model targets remain unchanged. This rule is applied to every vocabulary-rank readout in the baseline comparison. Dataset labels are not rewritten. LRE faithfulness still requires its exact top-1 token to match the model's top-1; the alternative wordings modify the lens axis.

Year tokenization. The answer boundary is checked with each actual tokenizer and the joint prompt–answer string. Where necessary, a trailing space is consumed in the relation prompt so that the next token is content. A standalone first answer token `1` is distinguished from multi-digit tokens such as `178`. Under the relation-level rule used for the primary comparison, PRESIDENTS_BIRTH_YEAR and PRESIDENTS_ELECTION_YEAR are excluded for Mistral, Gemma and both Qwen models, leaving 26 scored relations and 9,709 rows each. The artifact occurs on all 19 birth-year rows and 16 of the 19 election-year rows in each affected model. The relation-level rule removes both complete year relations, including the other three election-year rows, from the primary correlation. The rule does not depend on measured faithfulness. OLMo, GPT-J and Pythia retain both year relations and all 9,747 rows. All original rows and measurements remain available; the all-28 column in Table 3 shows the year-inclusive sensitivity.

Continent references. We retain the original seeded references for LANDMARK_ON_CONTINENT; there is no diversity correction in these reported values. Their selection and category counts are stored with each measurement. This setting is applied consistently to both methods and all models.

Leading newline and high norms. GPT-J 6B, Pythia 6.9B, Qwen3 8B Base and Qwen3.8 27B use `\n{} is known for`. A dagger marks these models in the result tables. OLMo, Mistral and Gemma use `{} is known for`. The newline addresses unusually large subject-state norms when content occupies the first token position. Table 2 reports the additional full-row GPT-J and Pythia ablations. Qwen's previously adopted correction is retained. The ablations change neither the LRE outcome nor its target tokens.

Numerical checks and provenance. Measurements use pinned checkpoint revisions, `float32`, eager attention, and the architecture's actual final normalization and output head. Scalar and batched Jacobian–vector products, finite differences, and cached computations are numerically checked. The report validates complete row multisets, duplicate counts, target IDs, reference counts, and source hashes, and recomputes aggregates from per-row ranks. Current numbers are generated from these records rather than copied from historical tables. Earlier shared-split, capped, correctness-filtered, and separate small-cohort numerical analyses are archived outside the compiled paper and are not used to substantiate its current claims.

Table 2: Full-evaluation newline ablation for the two additional models. All 28 relations, 9,747 rows, LRE results and target tokens are fixed.

Model	Original ρ	Newline ρ	Original norm	Newline norm
GPT-J 6B	0.1883	0.4346	4209	123
Pythia 6.9B	0.1746	0.5027	2876	282

Norm columns give median subject-state norms for rows whose subject endpoint was at token position zero before adding the newline (142 GPT-J rows; 83 Pythia rows). The newline shifts the subject away from position zero without changing its content token IDs. Qwen3 8B Base and Qwen3.8 27B retain the previously adopted newline correction; their current full-evaluation correlations are in Table 1.

Table 3: Correlation sensitivity to LRE parameter selection and year scoring. All columns use complete relation evaluations.

Model	Primary n	Selected	Frozen	All 28
OLMo 3 7B	28	0.7355	0.7046	0.7355
GPT-J 6B†	28	0.4346	0.4609	0.4346
Pythia 6.9B†	28	0.5027	0.4026	0.5027
Mistral 7B	26	0.6561	0.5986	0.3687
Gemma 2 9B	26	0.6889	0.6862	0.4541
Qwen3 8B Base†	26	0.6647	0.6490	0.4147
Qwen3.8 27B†	26	0.5539	0.5296	0.3037

Selected: best layer/scaling on full evaluation. Frozen: original layer/scaling retained, outcomes evaluated on all rows. All 28: both year relations included as a tokenization sensitivity, using selected LRE parameters. †Newline models as in Table 1.

B BASELINE COMPARISONS

Readout definitions. The contrastive/differential Jacobian lens (CEL) decodes $\mathrm{mean}_i\, J_i(s_e - s_i)$; the noncontrastive/raw Jacobian lens (NEL) decodes $\mathrm{mean}_i\, J_i s_e$. Logit lens decodes s_e, and differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Every readout applies the same final normalization and output head. Scores are the negative median of $\log_{10}(\mathrm{rank}+1)$ for zero-based ranks, with the gender rule above. Every eligible relation uses every original row.

Attribute rate. Following Geva et al. (2023), we compare filtered top-50 logit-lens predictions with attributes retrieved from Wikipedia. The retrieval queries cover every original subject, including those used as demonstrations or references. The source length, stopword and retrieval filters are retained. If no prediction survives the filter, the original overlap fraction is undefined; for this full-row comparison we explicitly extend it to zero and retain the row. The accompanying measurements record every such case. This convention changes the metric for those cases and avoids a model-specific filtered cohort.

B.1 OLMO CORPUS-FREQUENCY BASELINES

A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. The saved subject, object and joint counts cover every original fact string. We restore original duplicate multiplicities and recompute relation aggregates before correlating them with the current LRE outcomes. Unobserved conditioning marginals receive the explicit zero convention noted in the table, and no row is dropped.

These associations remain exploratory: relations can share subjects and objects, the counts do not resolve aliases, and p-values are unadjusted. They are not evidence that training frequency causes the observed readout or LRE performance.

Table 4: Neural and Wikipedia attribute baselines using the same full evaluations, targets, reference sets, layer, prompt correction, gender alternatives and year rule as the contrastive lens.

Model	CEL	NEL	Logit	Diff. logit	Attribute
OLMo 3 7B	0.7355	0.7650	0.2056	0.4390	-0.4016
GPT-J 6B†	0.4346	0.4196	0.4669	0.2020	-0.1035
Pythia 6.9B†	0.5027	0.5063	-0.2047	0.1845	0.0041
Mistral 7B	0.6561	0.6711	0.6581	0.5186	-0.2202
Gemma 2 9B	0.6889	0.6868	0.4728	0.3436	-0.5515
Qwen3 8B Base†	0.6647	0.6232	0.7908	0.5952	-0.0557
Qwen3.8 27B†	0.5539	0.5717	0.3404	0.4492	-0.0099

CEL: contrastive/differential Jacobian lens; NEL: raw/noncontrastive Jacobian lens. Higher scores mean better target ranks. †Leading newline. Attribute overlap is zero when no top-50 predictions survive the published length/stopword filter; no evaluation rows are dropped. These are point estimates, not evidence of significant differences between readouts.

Table 5: Attribute-rate coverage under the full primary evaluation.

Model	Scored rows	Empty filtered predictions
OLMo 3 7B	9,747	0
GPT-J 6B†	9,747	0
Pythia 6.9B†	9,747	2,663
Mistral 7B	9,709	0
Gemma 2 9B	9,709	1
Qwen3 8B Base†	9,709	0
Qwen3.8 27B†	9,709	0

Rows with no surviving top-50 prediction are assigned zero overlap in the reported extension. No original row is removed for an undefined attribute rate.

Table 6: Anchored neutral Jacobian readout at fixed $\beta = 1$, under the same full evaluation and scoring corrections.

Model	Spearman ρ
OLMo 3 7B	0.4070
GPT-J 6B†	0.4548
Pythia 6.9B†	0.2282
Mistral 7B	0.3197
Gemma 2 9B	0.4044
Qwen3 8B Base†	0.4229
Qwen3.8 27B†	0.4537

The readout averages $f(s_i) + J_i(s - s_i)$ under the neutral prompt. The LRE outcome still uses its selected relation-context layer and scaling. No cross-relation tuning of the neutral readout is performed.

Table 7: OLMo corpus-frequency baselines, reaggregated over all 9,747 original rows and correlated with full-evaluation LRE faithfulness across all 28 relations.

Feature	Spearman ρ	p
Mean subject document count	0.1380	0.484
Mean object document count	0.5712	0.001
Mean joint document count	0.4568	0.015
Mean $\log(1 + c_{so})$	0.4191	0.026
$\log(1 + \text{mean}(c_{so}))$	0.4568	0.015
Mean $P(o \mid s)$	0.4486	0.017
Mean $P(s \mid o)$	−0.4149	0.028
Mean $P(o \mid s) - P(s \mid o)$	0.4826	0.009
Micro $P(o \mid s)$	0.5504	0.002
Micro $P(s \mid o)$	−0.1453	0.461

Counts use binary document presence across the three saved OLMo training-mix scans. Duplicate dataset rows receive their original multiplicity. An unobserved conditioning marginal gives a conditional feature of zero, rather than dropping the row; 207 rows have at least one unobserved marginal. Micro conditionals divide summed joint counts by summed marginal counts. Values are exploratory and p-values are unadjusted.

Table 8: Fixed lens layers and LRE candidate layers (one-based block outputs).

Model	Lens layer	LRE candidate layers
OLMo 3 7B	17	5, 11, 14, 17, 20, 23
GPT-J 6B†	15	4, 10, 12, 15, 18, 20
Pythia 6.9B†	17	5, 11, 14, 17, 20, 23
Mistral 7B	17	5, 11, 14, 17, 20, 23
Gemma 2 9B	22	7, 14, 18, 22, 26, 30
Qwen3 8B Base†	19	6, 12, 16, 19, 22, 26
Qwen3.8 27B†	34	10, 22, 28, 34, 40, 46

Every LRE layer is crossed with $\beta \in \{0.5, 1, 1.5, 2, 3, 5\}$. The lens layer is fixed across relations within each model. †Leading newline.

C JACOBIAN–VECTOR PRODUCTS

Neither the operator nor the lens ever needs J_i itself, only products $J_i u$: for the lens this is explicit in Eq. 5, and for the LRE it is what the expansion in Eq. 3 buys, since it turns $W = \mathrm{mean}_i\, J_i$ into a sum of such products. We obtain each one by the standard double-backward construction (Pearlmutter, 1994): differentiate $\langle u', f(h)\rangle$ with respect to h with u' held as a free variable, then differentiate the result with respect to u' in the direction u. Each product therefore costs a constant number of forward and backward passes rather than the d_{model} passes a materialized Jacobian would need. The construction requires `float32` and eager attention, as the fused attention kernels are not twice differentiable.

D LAYER SELECTION

The lens uses one fixed intermediate layer per model, with no per-relation layer tuning. The LRE sweeps the candidate layers shown in Table 8. This does not establish robustness to layer choice; a systematic layer-sensitivity study is left to future work.

E ALL OLMO RELATIONS

F FULL-ROW FACT CORRELATIONS

The ranks are paired by subject, object and duplicate occurrence. Each rank is relative to that method's own relation-context target; the last column reports how often those targets differ.

Table 9: OLMo full-evaluation results for all 28 original relations. k is rows per distinct full object; A_r is the contrastive lens score.

Relation	Rows	k	Faithfulness	A_r
country_language	24	1.7	1.0000	−1.2205
name_birthplace	31	3.9	0.9677	−1.3222
name_religion	31	6.2	0.9677	−1.3979
person_plays_pro_sport	318	63.6	0.9528	−1.2304
landmark_on_continent	947	189.4	0.9229	−2.4265
country_largest_city	24	1.0	0.9167	−0.8116
person_native_language	919	30.6	0.9151	−1.6902
person_plays_instrument	513	85.5	0.8908	−1.7160
city_in_country	27	1.3	0.8889	−0.4771
country_capital_city	24	1.0	0.8750	−0.8451
person_band_lead_singer	21	1.0	0.8571	−1.4914
landmark_in_country	836	9.2	0.8014	−1.6946
presidents_election_year	19	1.1	0.7895	−3.4478
presidents_birth_year	19	1.3	0.7368	−4.4379
person_occupation	821	26.5	0.7113	−2.7364
product_by_company	522	17.4	0.6801	−1.8750
person_plays_position_in_sport	952	68.0	0.6071	−3.6421
person_university	91	1.3	0.5934	−4.3814
company_hq	674	4.1	0.5445	−4.2339
name_gender	19	9.5	0.5263	−1.0414
superhero_person	100	1.0	0.5000	−2.6212
star_constellation	362	11.7	0.4696	−3.5048
country_currency	30	1.3	0.4000	−4.4923
pokemon_evolutions	44	1.0	0.3182	−3.4528
superhero_archnemesis	96	1.1	0.3021	−3.8968
person_father	991	1.0	0.0747	−3.7077
person_mother	994	1.0	0.0724	−4.1824
company_ceo	298	1.0	0.0537	−4.5027

Every original row and duplicate is retained. Eight demonstration and eight reference rows are also part of the evaluation. Both year relations are retained for OLMo after token-boundary auditing.

Table 10: Full-row OLMo rank correlations within each relation, with target-context disagreement counts.

Relation	Rows	Spearman ρ	Different targets
city_in_country	27	0.163	1
company_ceo	298	0.073	77
company_hq	674	0.212	158
country_capital_city	24	0.576	0
country_currency	30	0.280	2
country_language	24	undefined	2
country_largest_city	24	0.317	0
landmark_in_country	836	0.424	92
landmark_on_continent	947	0.305	47
name_birthplace	31	0.072	2
name_gender	19	0.235	0
name_religion	31	−0.112	1
person_band_lead_singer	21	0.415	0
person_father	991	0.060	243
person_mother	994	0.087	227
person_native_language	919	0.288	22
person_occupation	821	−0.388	262
person_plays_instrument	513	−0.035	264
person_plays_position_in_sport	952	0.284	493
person_plays_pro_sport	318	0.281	22
person_university	91	−0.401	20
pokemon_evolutions	44	0.583	2
presidents_birth_year	19	−0.123	0
presidents_election_year	19	−0.171	2
product_by_company	522	0.218	99
star_constellation	362	0.337	29
superhero_archnemesis	96	0.310	42
superhero_person	100	0.664	6